

Erdős Problem 193: proof skeleton

The construction gives an unconditional negative answer: there is an infinite finite-step walk in $\mathbb{Z}^3$ with no three collinear vertices.

- 1. Build the nested Hilbert path and record its terminal state.** Decode an even-padded base-4 word $w = q_{\ell-1} \cdots q_0$. The symbol $g \in K$ is the mutable current decoder state, initialized as $g = I$; it is one of four maps on coordinate bits:

$$\begin{aligned} I(x, y) &= (x, y), & S(x, y) &= (y, x), \\ T(x, y) &= (1 - y, 1 - x), & C(x, y) &= (1 - x, 1 - y), \end{aligned} \quad K = \{I, S, T, C\}.$$

Digit $q \in \{0, 1, 2, 3\}$ has a child corner c_q and a transition map $r_q \in K$:

q	0	1	2	3
c_q	(0, 0)	(0, 1)	(1, 1)	(1, 0)
r_q	S	I	I	T

Here “update” means replacement of the current orientation map: when digit q_j is read, emit

$$(x_j, y_j) = g(c_{q_j})$$

and then set $g \leftarrow g \circ r_{q_j}$. Thus r_q is a digit-indexed map, unrelated to the integer r used in Steps 5–7. Finally set

$$H_\ell(w) = \left(\sum_{j=0}^{\ell-1} x_j 2^j, \sum_{j=0}^{\ell-1} y_j 2^j \right),$$

After the last digit q_0 , define $\sigma(w)$ to be the resulting value of g . Thus g denotes the changing intermediate state, while $\sigma(w)$ denotes its terminal value. The four-copy induction shows that H_ℓ is bijective, consecutive inputs are lattice neighbors, and two leading zero digits change neither H_ℓ nor σ . Hence they define an injective infinite path

$$H : \mathbb{N} \rightarrow \mathbb{N}^2, \quad \|H(n+1) - H(n)\|_1 = 1,$$

together with a terminal state $\sigma(n) \in K$.

- 2. Define the planar two-adic invariant.** For $u = (u_x, u_y) \neq 0$, put

$$\begin{aligned} p(u) &= \min\{\nu_2(u_x), \nu_2(u_y)\}, \\ \epsilon(u) &= \begin{cases} 1, & \nu_2(u_x) = \nu_2(u_y), \\ 0, & \nu_2(u_x) \neq \nu_2(u_y), \end{cases} \quad V(u) = 2p(u) + \epsilon(u). \end{aligned}$$

Multiplying both coordinates by a positive integer k gives

$$V(ku) = V(u) + 2\nu_2(k).$$

- 3. Prove the same-state pair law.** Let $m \neq n$ with $\sigma(m) = \sigma(n)$, and let j be the least base-4 place where their even-padded words differ. Equal terminal states let us reverse their common lower-digit suffix to one common state immediately after place j . The corners c_0, c_1, c_2, c_3 form a cyclic Gray code, so at binary place j exactly one coordinate differs when the two base-4 digits differ oddly, and both differ when their difference is 2 (mod 4). Therefore

$$V(H(n) - H(m)) = \begin{cases} 2j, & \nu_2(|n - m|) = 2j, \\ 2j + 1, & \nu_2(|n - m|) = 2j + 1, \end{cases}$$

and hence

$$V(H(n) - H(m)) = \nu_2(|n - m|).$$

- 4. Select bounded-gap indices with one terminal state.** Use

$$\rho(I) = 5, \quad \rho(S) = 1, \quad \rho(T) = 13, \quad \rho(C) = 3,$$

corresponding respectively to suffixes $11_4, 01_4, 31_4, 03_4$, and define

$$n_a = 16a + \rho(\sigma(a)).$$

Each suffix contributes a second copy of the prefix state, so every selected word ends in

$$\sigma(n_a) = I.$$

Since $\rho(K) = \{1, 3, 5, 13\}$,

$$4 \leq n_{a+1} - n_a \leq 28.$$

5. Reduce a hypothetical collinear triple to one common planar vector. On any set $E \subseteq \mathbb{N}$ where σ is constant, lift by

$$Q(n) = (H(n), n).$$

Suppose $a < b < c$ in E give three collinear lifts. Write

$$A = b - a, \quad B = c - b, \quad X = H(b) - H(a), \quad Y = H(c) - H(b).$$

Collinearity of (X, A) and (Y, B) gives $BX = AY$. If

$$A = gr, \quad B = gs, \quad g = \gcd(A, B), \quad \gcd(r, s) = 1,$$

coordinatewise divisibility gives a nonzero $W \in \mathbb{Z}^2$ such that

$$X = rW, \quad Y = sW.$$

6. The adjacent pairs force both reduced gaps to be odd. Apply the pair law and scaling identity to (a, b) and (b, c) :

$$\nu_2(g) + \nu_2(r) = V(W) + 2\nu_2(r),$$

$$\nu_2(g) + \nu_2(s) = V(W) + 2\nu_2(s).$$

Thus $\nu_2(r) = \nu_2(s)$. Since $\gcd(r, s) = 1$, both valuations are zero, so r, s are odd and

$$V(W) = \nu_2(g).$$

7. The endpoint pair gives the contradiction. Since

$$c - a = g(r + s), \quad H(c) - H(a) = (r + s)W,$$

put $t = \nu_2(r + s)$. The pair law and scaling identity for (a, c) give

$$\nu_2(g) + t = V((r + s)W) = V(W) + 2t = \nu_2(g) + 2t.$$

Hence $t = 0$, so $r + s$ is odd. But Step 6 made both r and s odd, so their sum is even. Contradiction. Therefore every constant- σ lift, in particular the selected lift, has no three collinear points.

8. Form the required finite-step walk. Define

$$P_a = Q(n_a) = (H(n_a), n_a).$$

It is infinite because its third coordinate strictly increases. For

$$P_{a+1} - P_a = (d_x, d_y, d_z),$$

the bounded selector gaps and unit Hilbert steps give

$$4 \leq d_z \leq 28, \quad |d_x| + |d_y| \leq d_z.$$

Thus every step lies in the fixed finite set

$$S = \{(d_x, d_y, d_z) \in \mathbb{Z}^3 : 4 \leq d_z \leq 28, |d_x| + |d_y| \leq d_z\}.$$

Step 7 excludes collinear triples, so $(P_a)_{a \geq 0}$ is the required counterexample.